

1	B	1 unit is 0.100m.
2	C	Component of initial velocity perpendicular to the wall is to the right. $v_{\perp i} = 4.0 \sin 30^\circ$ Component of final velocity perpendicular to the wall is to the left. $v_{\perp f} = 3.0 \sin 30^\circ$ Change in momentum and taking to the left positive, $\Delta v_{\perp} = v_{\perp f} - v_{\perp i} = 3.0 \sin 30^\circ - (-4.0 \sin 30^\circ) = 3.5 \text{ kg m s}^{-1}$
3	B	Air resistance is non-uniform in all directions as air resistance is dependent on